

HEALTHCONNECT CLINIC: ANALYTICAL TESTING, KPI VALIDATION & DASHBOARD REFINEMENT REPORT

Track: Data Analytics

Project: HealthConnect Clinic Experience Lab — Week 7

Document Type: Analytics Testing, Refinement & Validation Report

Submission Scope: Complete Week 7 Deliverable Package

Section 1: Week 6 > Week 7 Project Transition

1. Week 6 Baseline Context

- **Main Week 6 Deliverable:** Power BI Interactive Diagnostic Dashboard (HealthConnect_Dashboard_v1.pbix) and Week 6 Advanced Analytics Report.
- **Core Findings:** Validated non-attendance baseline at 48.5% No-Show Rate, 46.3% Attendance Rate, and 53.7% Wasted Capacity (1,210 Lost Provider Hours). Pinpointed compounding friction in patients traveling >15km with lead times >14 days (44.2%-57.0% drop-off).
- **Identified Limitation:** The Week 6 dashboard lacked systematic visual/DAX stress testing. Matrix visual granularities displayed raw decimal slices (15.10, 15.20), KPI titles created ambiguity (SMS No Show Rate % set to 0.46 decimal), and table visuals displayed non-sensical sums (38,016 lead days).

2. Week 7 Objective & Scope

Week 7 transitions our analytics output from initial integration into Systematic Testing, Refinement, and End-to-End Validation. Rather than rebuilding EDA, this phase stress-tests DAX calculation accuracy, evaluates visual usability, validates cross-track data alignment with Data Science, and re-tests refined deliverables.

Section 2: Testing Readiness & Systematic Test Plan

1. Component Under Test

- **Primary Component:** Power BI Interactive Analytics Dashboard (HealthConnect_Dashboard_v2.pbix) and underlying DAX measures.
- **Connected Tracks:** Data Science (Predictive Modeling) and Project Management (Dependency & Risk Tracking).

Section 3: KPI Validation & DAX Refinement

To ensure absolute mathematical integrity between the raw dataset and visual outputs, all underlying DAX measures were re-evaluated and hardened against blank records or erroneous aggregations.

1. Validated KPI Calculations

- **Overall No-Show Rate %:** 48.5
- **Attendance Rate %:** 46.3
- **Wasted Capacity %:** 53.7%
- **Lost Capacity Hours:** 1,212.5 Hours Formatted to 1.21K

2. Refined DAX Code Library

Code snippet

```
// 1. Overall No-Show Rate % (Hardened against nulls)
No Show Rate % =
VAR ValidRows = FILTER('HealthConnect_Appointment_Data',
NOT(ISBLANK('HealthConnect_Appointment_Data'[appointment_outcome])))
VAR NoShows = COUNTROWS(FILTER(ValidRows,
'HealthConnect_Appointment_Data'[appointment_outcome] = "No-Show"))
RETURN
DIVIDE(NoShows, COUNTROWS(ValidRows), 0)
```

```
// 2. Calculated Column: Distance Tier Categorization
Distance Tier =
SWITCH(
    TRUE(),
```

```

    'HealthConnect_Appointment_Data'[distance_to_clinic_km] < 5, "<5
km",
    'HealthConnect_Appointment_Data'[distance_to_clinic_km] <= 15, "5-
15 km",
    'HealthConnect_Appointment_Data'[distance_to_clinic_km] > 15, ">15
km",
    "Unknown"
)

// 3. Calculated Column: High-Risk Action Queue Flag
Needs Urgent Outreach =
IF(
    'HealthConnect_Appointment_Data'[booking_lead_days] > 14 &&
    'HealthConnect_Appointment_Data'[reminder_sent] = "No",
    "Needs Outreach",
    "Standard"
)

```

Section 5: HC-POD Cross-Track Testing & Validation

Mandatory cross-track testing was executed with the Data Science Track and Project Management Track to validate end-to-end coherence.

1. Data Science Track Collaboration

- **Project Dependency:** Alignment between Analytics segment findings and predictive model feature inputs.
- **Input Received:** Data Science Gradient Boosting candidate model confusion matrix (316 True Positives, 175 False Positives) and error analysis indicating long lead times drove false positive classifications.
- **Input Provided:** Binned risk tiers (Distance Tier > 15km and Lead Time > 14 days) and unconfirmed reminder status flags.
- **Testing Activity:** Re-tested whether filtering dashboard metrics for high-risk combinations aligned with Data Science's model sensitivity thresholds.

- **Retest Result & Impact:** Validated that setting a 0.45 prediction threshold in Data Science matches our administrative queue risk flag, catching 74% of potential no-shows for intervention.

2. Project Management Track Collaboration

- **Project Dependency:** Integration & Risk Register updates.
- **Testing Activity:** Logged DAX display bugs (TS-002, TS-004) into the central PM Issue Tracker and verified issue resolution following re-testing.

Section 6: Updated Business Insights & Recommendations

1. **Prioritize Lead Time >14 Days for Intervention:** Testing confirms booking lead time is the single strongest predictor of appointment failure across both analytics and machine learning models.
2. **Deploy Targeted Phone Outreach for Non-SMS Cohorts:** While SMS achieves a 45.8% drop-off rate, unreminded slots suffer 51.4% to 54.6% drop-offs. Administrative phone calls must be prioritized for patients over 65 living >10 km away.
3. **Automate Capacity Recovery:** Implement dynamic backfilling on unconfirmed appointments flagged in the refined Needs Urgent Outreach dashboard queue.

Section 7: Week 7 Concise Project Summary

- **What was Planned vs. Tested:** Planned to test dashboard DAX accuracy, visual usability, and feature alignment with Data Science. Systematically tested 6 core visual/DAX scenarios.
- **Key Weaknesses Found:** Raw decimal formatting on percentage cards, unreadable matrix columns, and invalid total row summaries in table visuals.
- **Refinements Made:** Created Distance Tier DAX column, reformatted KPI measure properties to percentages, removed table totals, and enabled strict visual cross-filtering.
- **Collaborating Track:** Data Science & Project Management.
- **Retest Outcome:** All 6 test scenarios passed 100% on re-test, locking in HealthConnect_Dashboard_v2.pbix for final presentation.

- **Week 8 Readiness:** Analytics components are fully validated, bug-free, and ready for final integration and presentation.